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Océane Boulais ; Daniel Schar; Josh Levy; Katherine Kim; Natalie Levy; Jessica Reichert; Nina Schiettekatte; Daniel Wangpraseurt; Josh Madin; Aaron Thode



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Animal Bioacoustics: Paper 1aAB21**A demonstration of acoustic enrichment in Hawai'i
using autonomous cameras for reef fish larval detection****Océane Boulais***Scripps Institution of Oceanography, UC San Diego: University of California San Diego, La Jolla, CA,
92037; oboulais@ucsd.edu***Daniel Schar***University of Hawai'i at Mānoa Hawaii Institute of Marine Biology, Honolulu, HI; schar@hawaii.edu***Josh Levy***Applied Research Laboratory, University of Hawai'i at Manoa, Honolulu, HI; josh.levy@arllcollab.onmicrosoft.com***Katherine Kim***Greeneridge Sciences, Goleta, CA; khkim@greeneridge.com***Natalie Levy***Department of Nanoengineering, University of California San Diego, La Jolla, CA; nrlevy@ucsd.edu***Jessica Reichert and Nina Schiettekatte***University of Hawai'i at Mānoa Hawaii Institute of Marine Biology, Honolulu, HI; jhaver@hawaii.edu;
nina.schiettekatte@gmail.com***Daniel Wangpraseurt***Department of Nanoengineering, University of California San Diego, La Jolla, CA; dwangpraseurt@ucsd.edu***Josh Madin***University of Hawai'i at Mānoa Hawaii Institute of Marine Biology, Honolulu, HI; jmadin@hawaii.edu***Aaron Thode***University of California San Diego Scripps Institution of Oceanography, La Jolla, CA; athode@ucsd.edu*

Reef fish and their cryptobenthic larvae are a fundamental aspect of a productive and healthy coral reef ecosystem. Playbacks of ambient sound recorded from healthy shallow coral reefs, known as “acoustic enrichment” (AE), have been demonstrated to attract fish larvae and potentially coral larvae to degraded or artificial reefs. Initial AE studies relied on frequent diver surveys for evaluating performance, but these approaches are labor intensive and impractical for larger-scale and more remote studies. Here, we use autonomous cameras to assess AE efficacy on artificial structure deployments, conducted between July and September 2023 in Kāne'ohe Bay (O'ahu, Hawai'i). The deployments consisted of one control and one treatment site 65m apart, at 5m water depth. To document ecosystem responses, both sites used open-source burst-mode autonomous cameras, a directional vector sensor seafloor acoustic recorder, and hydrophones. A 3.8 kW-hr NiMh battery supply permitted treatment playbacks from sunset to sunrise for three weeks. Burst-mode images successfully documented fish larvae presence and movement. Image processing methods allowed automated estimates of numbers of larvae in a frame, when larvae were identified during a manual review. The resulting analysis demonstrates that the AE site attracted at least 23 times more larvae than the control site.

INTRODUCTION

Fish perform a diverse array of functional processes on coral reefs, including consuming algae that overgrow coral¹. Previous research has demonstrated the presence of reef fish plays a crucial role in coral ecosystem rehabilitation^{2,3,4}. Different functional groups of reef fish, such as excavators, scrapers, browsers, grazers, and detritivores, contribute significantly to the ecological balance and recovery potential of coral reefs⁵. The persistence of reef fish populations is largely dependent on the process of recruitment, where juvenile fish, following their planktonic larval phase, utilize various sensory signals to locate, approach, and inhabit reef environments⁶. Yet, when reefs are in a state of decline, they emit less appealing olfactory and acoustic signals to young fish, leading to a decrease in fish settlement compared to healthy coral ecosystems⁷. Actively counteracting these sensory alterations associated with reef degradation could potentially increase the appeal of these habitats, encourage the settlement and permanence of crucial fish species through use of acoustic cues leveraged by settlement-stage fish, and bolster reef rehabilitation efforts at a local level^{8,9}.

Playbacks of ambient sound recorded from healthy shallow coral reefs, a technique known as acoustic enrichment (AE), has been demonstrated to attract a myriad of herbivorous reef fish to degraded or artificial reefs¹⁰. While those results were obtained from intensive and frequent diver surveys, applying and demonstrating AE on larger and more remote scales faces both power supply challenges and diver logistic restrictions, making sustained monitoring of local biological activity difficult. For example, a recent DARPA-funded program, called “REEFENSE”, seeks to enhance rapid coral ecosystem development on artificial structures to help protect Department of Defense coastal assets. A 100 m structure is eventually scheduled to be deployed offshore at a Marine Corps facility in O’ahu, Hawai’i, led by the Applied Research Laboratory at the University of Hawaii. However, access to the site by divers is limited by its remote location and distance from harbor facilities. Wave action at this open ocean-facing location also means that using solar panels to recharge the acoustic playback device is infeasible. As a way to address these challenges, the development of an autonomous means of both powering and evaluating AE protocols would contribute to the furthering of this monitoring effort.

Here, we demonstrate a trial AE experiment in Kāne’ohe Bay, O’ahu, using an autonomous playback device with three-week endurance and no surface expression. We show a capability of continuously monitoring the presence and movement patterns of mature reef fish and reef fish larvae at both a treatment and control site, using a burst photo mechanism aboard open-source cameras. We found that the automated camera can visually detect fish larvae, and subsequently, image processing techniques to count them in individual frames.

1. METHODS

A. ACOUSTIC ENRICHMENT EQUIPMENT

The design and implementation of an autonomous underwater acoustic playback system is centered around a programmable PIC-32 microcontroller. The system is powered by 180 D-cell nickel-metal hydride (NiMH) batteries in two pressure cases to achieve an energy storage capacity of 3.8 kW-hr, sufficient for three weeks of continuous nighttime operation. The batteries are stored inside two pressure cases with a length of 812.8 mm and diameter of 215.9 mm, which are deployed on the ocean floor side by side, with a 10 m cable connecting a Lubell LL916C underwater electrodynamic loudspeaker to the cases.

The system features a magnetic switch mechanism that allows for in-situ testing of the playback functionality. This switch can activate a six-minute playback test, independent of any pre-set

scheduling, providing a flexible and reliable testing framework for underwater acoustic studies. The playback software has a unique “playlist” for each day of the month. Each playlist lists the names of multiple WAV-formatted files. The software has a master schedule specifying start and stop times for each day of the month. When playback begins each evening, the playback device plays each file in the playlist sequentially, once the six-minute test file is completed. An internal log file notes the specific playlist file and times activated.

B. AUTONOMOUS CAMERAS

For continuous visual monitoring of the experiment, an open-source camera system was employed, based on the ESP32-CAM circuit board with 2MP resolution with a 1600 x 1200 at 15 frames per second image transfer rate and 65-degree field of view (FoV). A custom circuit board called a “Kilocam” (Ecologis, Inc.) uses an Arduino microcontroller to act as a scheduler and intervalometer for activating the ESP32-CAM image acquisition. The software was modified to permit a burst mode functionality, which allows the capture of images in rapid succession to effectively detect and document motion. To document reef fish and larvae, every two minutes the software acquired five images over roughly two seconds. The data acquisition is completely silent and non-invasive. The camera only operates during daytime hours, and the camera automatically adjusts its sensitivity based on lighting levels. Two packs of (6) AA 2400mAh 7.2V NiMH batteries are used to power both the ESP32-CAM and Kilocam, with everything encased in a custom PVC housing with acrylic windows. With its cost-efficiency and motion-capture capabilities through the burst mode, a camera permits extended monitoring of a location from sunrise to sunset during summer months in Hawai’i, averaging 13 hours of daylight operation time.

C. STUDY SITE

This work was located on the northwestern side of Moku o Lo‘e Island in Kāne‘ohe Bay at two sites (respectively, 21°26'16"N 157°47'33"W, 21°26'16"N 157°47'31"W), in close vicinity of the Hawai’i Institute of Marine Biology (HIMB) over flat sandy bottoms (Figure 1, 2, 3) at a depth of approximately 4 m.

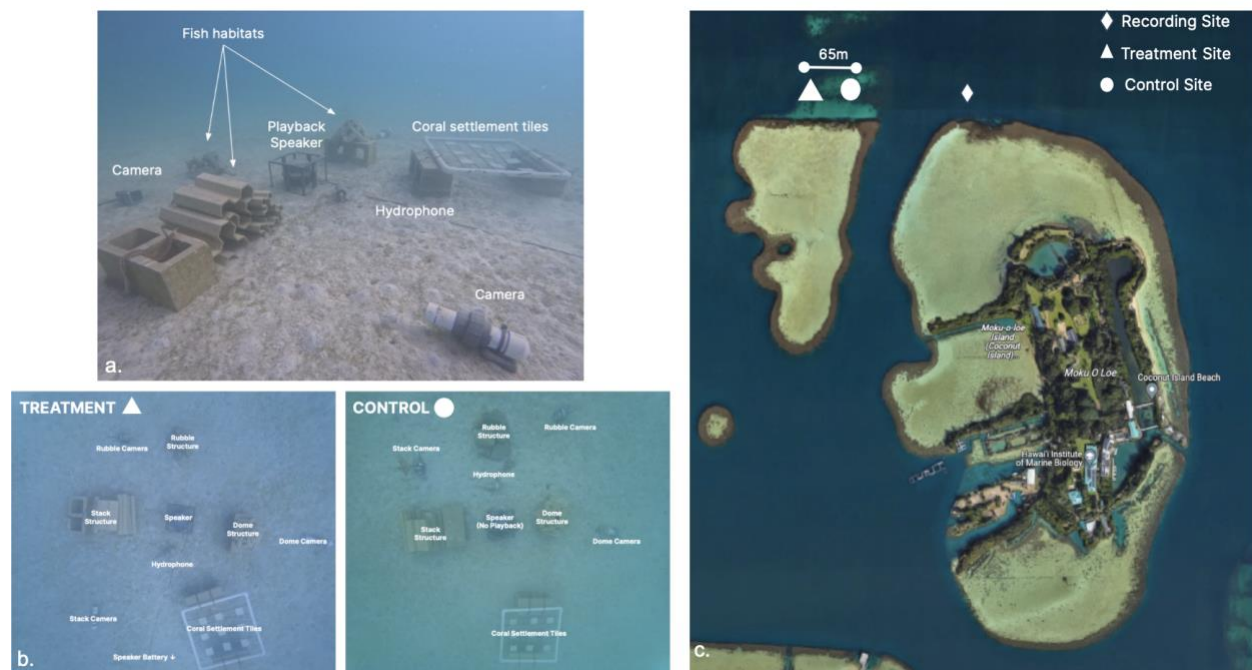


Figure 1: (a) Experimental layout of remote test site. Among the equipment visible are three ESP32-CAM autonomous cameras, a single ST300 Ocean Instruments hydrophone, three small fish settlement structures, a Lubell LL916C speaker, and coral settlement tiles. (b) Nadir view of the treatment and control sites during August 2023 deployments. Note the presence of a dummy speaker at the control site. (c) Locations of acoustic enrichment, treatment and control site, deployed approximately 65m apart at a reef site in Kāneʻohe Bay, Oʻahu, and monitored July and August 2023. The location where the playback files were recorded occurred off the northern side of Moku o Loʻe.

While one site served as a control location with no acoustic playbacks, a second site was placed 65 m away as the treatment site. The mean water depth at both locations was 4 m, and both locations were at least 10 m from nearby natural fringe reef outcroppings. The site had relatively little tourist traffic. In each study site, three distinct fish habitats were established: two 3D-printed artificial reef modules designed to enhance structural complexity, and a third habitat comprising a rubble pile, which was constructed using locally sourced crumbled concrete. One of the 3D-printed modules was a conical structure (40 cm diameter, 20 cm height) with openings in different sizes. The second structure was composed of three tube-shaped modules (on average, 23.5 cm height, 19.8 cm width) with different levels of complexity, stacked into a pyramid. Module designs were constructed in Blender v3.4, and printed with the Delta Wasp 40100 3D clay printer (Wasp, Italy) printing with clay (mid fire pottery clay, Soldate 60, Aardvark Clay and Supplies, United States). Printed modules were air-dried for ~10 days before firing in a kiln (Jupiter Sectional Kiln, L&L Kilns, United States).

D. ACOUSTIC ENRICHMENT PLAYBACK FILES

Passive acoustic data were collected by a SoundTrap ST 300 (Ocean Instruments, NZ; sensitivity 173.3 dB re 1 uPa/V) at 21°26'15"N 157°47'25"W in Kāneʻohe Bay during August 2022, where recorders continuously recorded healthy coral reef sound at 96 kHz sampling rate for eight days centered around the full moon phase, and an additional eight days centered around the new moon phase. These data were down sampled to 36 kHz WAV files using an FIR lowpass filter, trimmed to begin at sunset and end at sunrise, and transferred to the 128 Gb flash memory on the playback equipment.

This location was 300 m from the eventual test site. Acoustic playbacks were scheduled daily from 17:30 to 6:30, with each day featuring unique data with no repetition loop. The files chosen from 2022 were assigned to be broadcast back on days that matched the same lunar phase as the data of the original recording. Thus each day of an AE trial used a different recording. An example of a source playback file is depicted in Figure 2 where (a), shows that the coral reef ambient sound is dominated by crustacean snaps. An engineering deployment in July 2023 was used to adjust the playback source level so that the root mean square (RMS) received levels matched ambient rms levels (SNR 0) at 20 m range from the speaker. In order to measure the RMS level, a band range of 1-15 kHz was applied. Part (b) shows a sample of the same sound recorded at 20 m range in August 2023. As mentioned previously, the same seven-minute sound sequence was played every night before initiating the full playback, to facilitate verification that the playback device worked during post-deployment signal analysis.

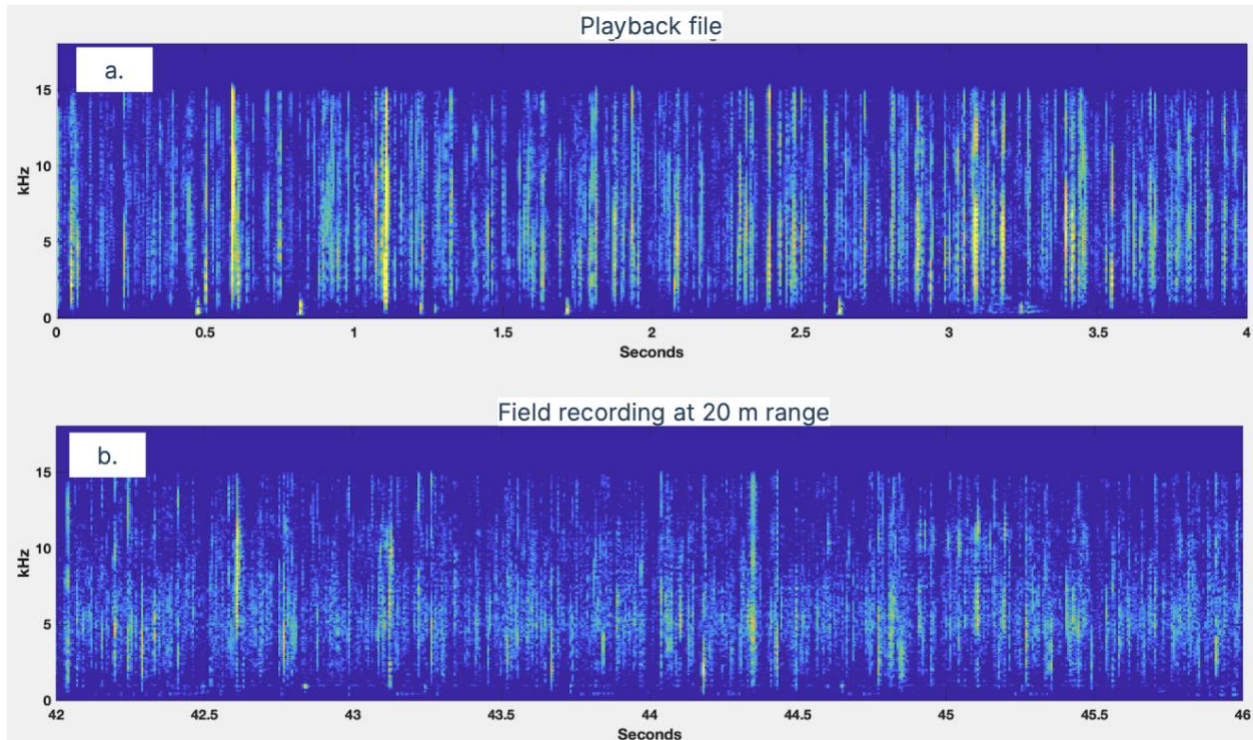


Figure 2: (a) shows that the coral reef ambient sound is dominated by crustacean snaps. (b) shows a sample of the same sound recorded at 20 m range in August 2023

3. RESULTS

The AE trial took place between August 12 and 24, 2023. Figure 1 shows the layout of both sites. Each site had three artificial fish attractant structures. Kilocams were exchanged every three days at each site. Between August 12 and 15 one camera was present at each site, focused on the “stack structure,” but after August 15 three cameras were placed at each site, with one camera assigned to each structure. One camera at the control site flooded between August 16 and 21, with the loss of all data. The cameras were placed about half a meter from each habitat, and swapped every three days. At the control site a dummy speaker was deployed in order to account for any visual attractive effects by the blue speaker. Acoustic data from the onsite ST300 Ocean Instruments hydrophones was collected alongside the visual data from the ESP32-CAM, to independently confirm playback success.

Despite a relatively low-resolution camera, the presence of fish larvae could be identified by a manual review. The burst images were invaluable for distinguishing the movement of larvae from passively drifting flocks in the water. The number of larvae visible in the images was large enough that bulk manual counts were impractical. A simple image segmentation algorithm was developed to count larvae in photo sequences that were flagged as having larvae. The algorithm subtracted two successive burst images from each other, highlighting objects that moved. After using a threshold to convert the output into a binary image, a morphological opening operation was used to restrict counts to objects between 5 and 100 pixels large. The number of objects detected was divided by two to yield the final larvae count in order to account for the “double-counted” fish larvae. Figure 3 shows an example of the image segmentation algorithm counting 82 larvae after two burst photos (< 1 second apart) are compared for relative changes via pixel subtraction. Occasional full manual counts of larvae were conducted to validate the performance of the bulk

counts. The final output metric is “MaxN”, or the maximum number of larvae counted in a single image over 30 minutes. The MaxN metric is particularly useful in situations where organisms are highly mobile and move in and out of the survey area, such as fish larvae, as it reduces the likelihood of overestimating population sizes due to repeated counts of the same individuals.

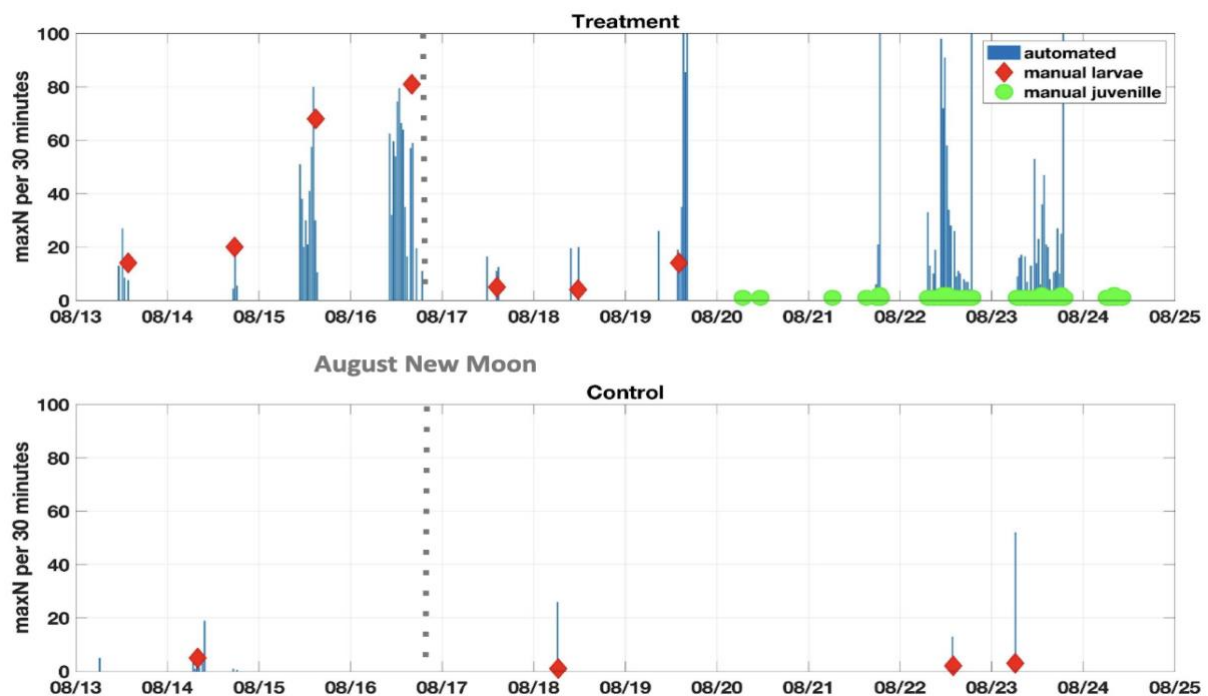
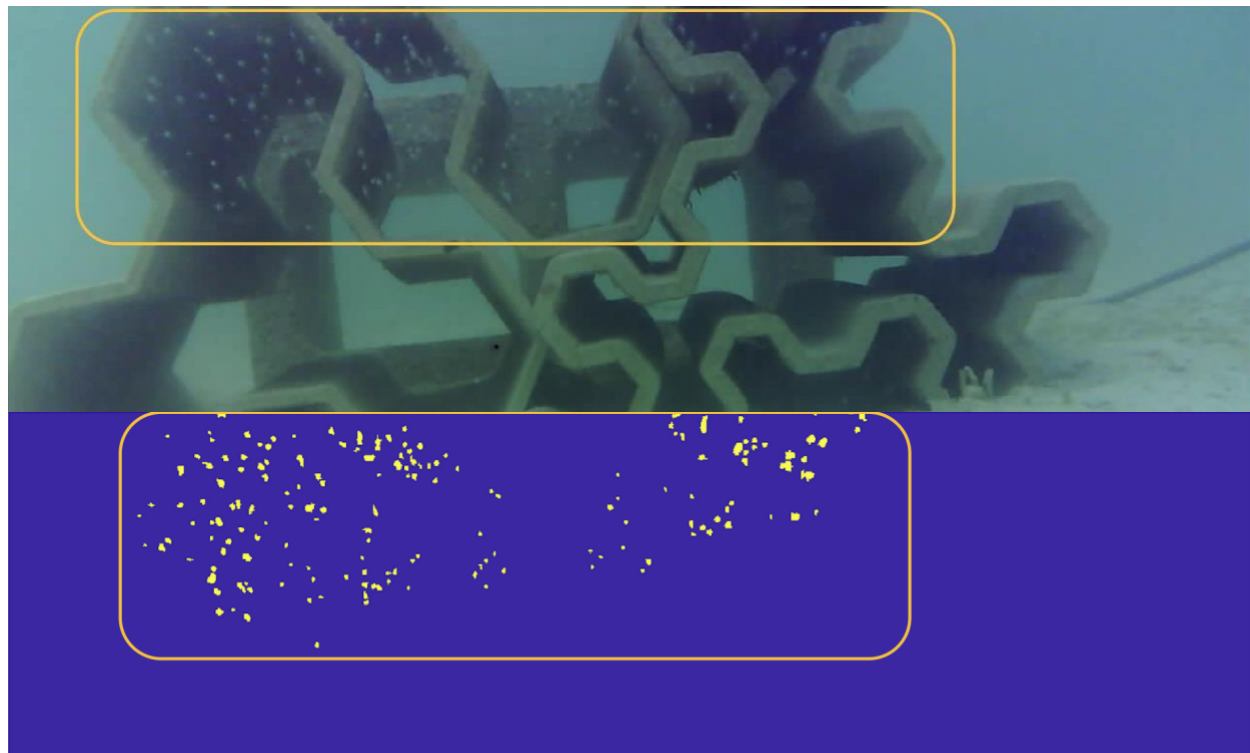


Figure 3: (top) Results of the reef fish larvae detection and counting algorithm and (middle) the source image collected by the autonomous camera. (bottom) shows a bar graph of the MaxN counts of fish larvae and juveniles at the treatment site (top) and control site (bottom). Red diamonds are the manual counts of larval fish, and green circles are manual counts of juvenile fish. The vertical blue bars are the automated larval MaxN produced by the image processing algorithm. The date of the new moon is shown by the vertical dashed line.

4. DISCUSSION

The presence of fish larvae observed between the treatment and control sites using autonomous underwater cameras differed strongly, with roughly 23 times larger MaxN values present at the treatment site than at the control site. The treatment site exhibited a consistent presence of numerous larvae and some juveniles over multiple days, contrasting sharply with the control site, where only a few larvae and no juveniles were detected. Notably, the largest counts occurred around the peak of the lunar new moon, as supported by existing literature¹⁰. While larvae counts dropped rapidly following the new moon, juveniles were spotted at the site over subsequent days. This could be due to specific life cycle traits and environmental adaptations of certain reef fish species, including a correlation between increased predator presence during dark nights which might lead to higher predation rates on larval fish^{11,12}.

Automated larval swarm counts seem consistent with the manual counts, particularly in situations when twenty or more larvae are present. However, this automation was less dependable for juvenile counts or in scenarios with sparse larval presence seen in Figure 3. Several external factors complicate the interpretation of the results, including shifts in camera coverage, a diver's site visit on August 16, and the failure of one camera at the control site between August 16th and 21th. The camera failure is the most serious situation, as the time covered at the control site was during the peak of the full moon. However, Figure 3 shows that the contrast in larvae presence is visible on the 14th and 15th, before the camera failure.

5. CONCLUSION

This work found that autonomous cameras can be used to evaluate acoustic enrichment of coral reefs without requiring intensive and invasive diver surveys. We also found that simple image processing methods could accurately estimate larvae counts in images, particularly when the images are known beforehand to contain fish larvae. While the current image processing methods are good for counting, and struggle with unsupervised detection, there exists capacity for optimization leveraging convolutional neural network architectures such as Faster R-CNN and Grid R-CNN in the future¹³. Fish larvae were confirmed to have a persistent presence during the summer new moon phase trial of an acoustic enrichment trial, with fish counts up to 23 times higher than observed at the control.

We conclude that autonomous cameras may be viable alternatives to diver surveys in remote and long-term AE studies, especially when attempting to document presence-sensitive marine organisms such as fish larvae, but that further efforts to automate image analysis are warranted. Our next research steps include the addition of lights to cameras for nighttime surveillance and deployments of cameras over longer intervals.

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Author Declarations

Conflict of Interest statement: The authors declare no competing or conflictual interests.

Data Availability: Data will be made available as requested to the corresponding author.